Supplementary Material: Traits predict forest phenological responses to photoperiod more than temperature

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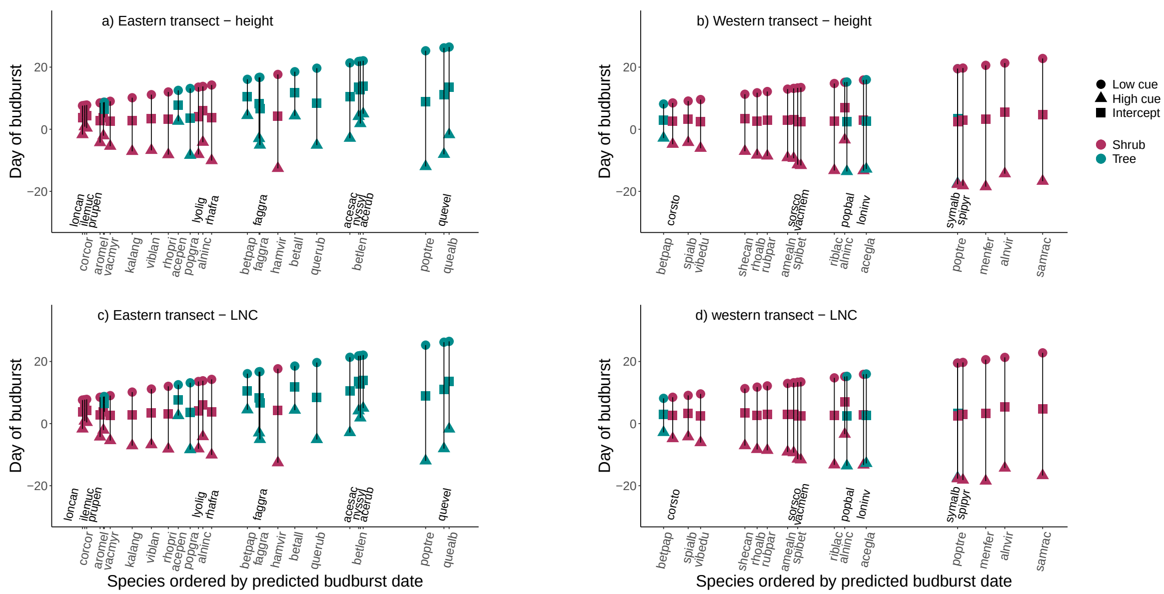

Figure S1: We found the estimated day of budburst differed between our full model (intercept plus cues, depicted as triangles for high cues and as circles for low cues), versus the intercepts only model (without cues, shown as squares). Species are ordered in increasing budburst dates for both the (a, c) eastern (b, d) and western populations, and in general span from early budbursting shrubs shown in red, to late budbursting trees shown in blue. (a, b) For traits such as height we found distinct partitioning of budburst across shrub and tree species, but this was not the case for all traits, (c, d) with the ranked order of species in our model of leaf nitrogen content exhibiting a highly mixed budburst order for our shrub and tree species.

Table S1: List of species names and abbreviations, as well as the transect in which each species was sampled from and plant architecure.

Species	Spp. abbreviation	Transect	Plant architecture
Acer glabrum	acegla	Western	Tree
Acer pensylvanicum	acepen	Eastern	Tree
Acer rubrum	acerub	Eastern	Tree
Acer saccharum	acesac	Eastern	Tree
Alnus incana	alninc	Eastern and Western	Shrub
Alnus viridis	alnvir	Western	Shrub
Amelanchier alnifolia	amealn	Western	Shrub
Aronia melanocarpa	aromel	Eastern	Shrub
Betula alleghaniensis	betall	Eastern	Tree
Betula lenta	betlen	Eastern	Tree
Betula papyrifera	betpap	Eastern	Tree
Betula papyrifera	betpap	Eastern and Western	Shrub
Corylus cornuta	corcor	Eastern	Shrub
Cornus stolonifera	corsto	Western	Shrub
Fagus grandifolia	faggra	Eastern	Tree
Fraxinus nigra	franig	Eastern	Tree
Hamamelis virginiana	hamvir	Eastern	Shrub
Ilex mucronata	ilemuc	Eastern	Shrub
Kalmia angustifolia	kalang	Eastern	Shrub
Lonicera canadensis	loncan	Eastern	Shrub
Lonicera involucrata	loninv	Western	Shrub
Lyonia ligustrina	lyolig	Eastern	Shrub
Menziesia ferruginea	menfer	Western	Shrub
Nyssa sylvatica	nyssyl	Eastern	Tree
Populus balsamifera	popbal	Western	Tree
Populus grandidentata	popgra	Eastern	Tree
Populus tremuloides	poptre	Eastern and Western	Shrub
Prunus pensylvanica	prupen	Eastern	Tree
Quercus alba	quealb	Eastern	Tree
Quercus rubra	querub	Eastern	Tree
Quercus velutina	quevel	Eastern	Tree
Rhamnus frangula	rhafran	Eastern	Shrub
Rhododendron albiflorum	rhoalb	Western	Shrub
Rhododendron prinophyllum	rhopri	Eastern	Shrub
Ribes lacustre	riblac	Western	Shrub
Rubus parviflorus	rubpar	Western	Shrub
Sambucus racemosa	samrac	Western	Shrub
Shepherdia canadensis	shecan	Western	Shrub
Sorbus scopulina	sorsco	Western	Shrub
Spiraea alba	spialb	Western	Shrub
Spiraea betulifolia	spibet	Western	Shrub
Spiraea pyramidata	spipyr	Western	Shrub
Symphoricarpos albus	symalb	Western	Shrub
Vaccinium membranaceum	vacmem	Western	Shrub
Vaccinium myrtilloides	vacmyr	Eastern	Shrub
Viburnum cassinoides	vibcas	Eastern	Shrub
Viburnum edule	vibedu	Western	Shrub
Viburnum lantanoides	viblan	Eastern	Shrub

Table S2: We obtained the temperature data we used to calculate field chilling from the nearest weather station to each site. Temperature data for the fall of 2019 and winter 2020 was obtained from the Government of Canada's Environmental and natural resources' historical data, using the Hope Slide weather station for our E.C. Manning park samples and the Smithers airport weather station for our Smithers samples (Government of Canada: Environment and natural resources, 2026). For our eastern samples, weather data for the fall of 2014 and winter 2015 was obtained from weather stations at Harvard Forest (Boose et al., 2026) and in the Government of Canada's Environmental and natural resources' historical data from the St. Hippolyte weather station (Government of Canada: Environment and natural resources, 2026). Listed below are the total chill portions that include both field and experimental chilling. Chill portions were calculated using the Dynamic Model in the chillR package (v. 0.73.1, Luedeling, 2020).

Site	Chilling treatment	Chill portions
Harvard forest	Field chilling	56.62
Harvard forest	Field chilling + 30 d at 4°C	94.06
St. Hippoltye	Field chilling	44.63
St. Hippoltye	Field chilling + 30 d at 4°C	82.06
Smithers	Field chilling + 21 d at 4°C	54.95
Smithers	Field chilling + 56 d at 4°C	74.67
Manning park	Field chilling + 21 d at 4°C	55.09
Manning park	Field chilling + 56 d at 4°C	75.33

Table S3: Summary output from a joint Bayesian model of height and budburst phenology in which species are partially pooled. The model includes environmental cues as z-scored continuous variables, allowing comparisons to be made across cues.

	Mean	5%	25%	75%	95%
Transect	7.58	-1.08	4.05	11.08	16.16
Latitude	0.05	-0.03	0.02	0.08	0.12
Transect x latitude	-0.20	-0.39	-0.28	-0.12	-0.01
Forcing	-10.31	-12.59	-11.09	-9.41	-8.35
Chilling	-13.44	-17.22	-14.81	-11.99	-9.89
Photoperiod	-2.54	-4.03	-3.20	-1.94	-0.81
Trait x forcing	0.28	-0.07	0.14	0.42	0.61
Trait x chilling	0.25	-0.46	-0.04	0.54	0.96
Trait x photoperiod	-0.34	-0.52	-0.41	-0.27	-0.16

Table S4: Summary output from a joint Bayesian model of diameter and budburst phenology in which species are partially pooled. The model includes environmental cues as z-scored continuous variables, allowing comparisons to be made across cues

	Mean	5%	25%	75%	95%
Transect	-24.13	-42.56	-31.70	-16.66	-5.88
Latitude	0.16	0.00	0.09	0.22	0.31
Transect x latitude	0.52	0.11	0.35	0.68	0.92
Forcing	-9.14	-11.41	-10.07	-8.30	-6.53
Chilling	-12.52	-16.21	-14.03	-11.12	-8.55
Photoperiod	-3.79	-5.68	-4.46	-3.08	-2.01
Trait x forcing	0.20	-0.06	0.09	0.30	0.46
Trait x chilling	0.16	-0.34	-0.04	0.37	0.65
Trait x photoperiod	-0.20	-0.34	-0.26	-0.15	-0.07

Table S5: Summary output from a joint Bayesian model of branch wood density and budburst phenology in which species are partially pooled. For this model the trait data was rescaled by 10 and environmental cues were included as z-scored continuous variables, allowing comparisons to be made across cues.

	Mean	5%	25%	75%	95%
Transect	7.24	4.42	6.04	8.42	10.01
Latitude	0.04	0.01	0.02	0.05	0.06
Transect x latitude	-0.15	-0.21	-0.17	-0.12	-0.09
Forcing	-14.91	-22.87	-17.59	-11.74	-8.63
Chilling	-20.94	-33.15	-25.21	-16.26	-9.75
Photoperiod	-0.61	-4.12	-2.14	0.82	3.30
Trait x forcing	1.73	-0.26	0.89	2.53	3.75
Trait x chilling	2.66	-0.94	1.32	4.04	6.11
Trait x photoperiod	-0.93	-1.99	-1.38	-0.51	0.19

Table S6: Summary output from a joint Bayesian model of leaf mass area and budburst phenology in which species are partially pooled. For this model the trait data was rescaled by 100 and environmental cues were included as z-scored continuous variables, allowing comparisons to be made across cues.

	Mean	5%	25%	75%	95%
Transect	-22.04	-26.60	-23.92	-20.18	-17.51
Latitude	-0.15	-0.20	-0.17	-0.14	-0.11
Transect x latitude	0.48	0.38	0.44	0.52	0.58
Forcing	-2.68	-19.52	-9.07	4.01	13.94
Chilling	-2.67	-30.26	-13.87	8.06	26.04
Photoperiod	-13.97	-23.12	-18.10	-10.12	-3.54
Trait x forcing	-0.59	-1.97	-1.15	-0.04	0.83
Trait x chilling	-0.87	-3.29	-1.80	0.09	1.43
Trait x photoperiod	0.89	0.01	0.57	1.24	1.65

Table S7: Summary output from a joint Bayesian model of leaf nitrogen content and budburst phenology in which species are partially pooled. The model includes environmental cues as z-scored continuous variables, allowing comparisons to be made across cues.

	mean	5%	25%	75%	95%
Transect	-4.71	-12.82	-8.03	-1.40	3.24
Latitude	-0.14	-0.21	-0.17	-0.11	-0.06
Transect x latitude	0.04	-0.13	-0.03	0.11	0.22
Forcing	-14.43	-32.60	-21.41	-7.56	3.76
Chilling	-35.11	-68.11	-47.92	-22.15	-4.07
Photoperiod	6.39	-2.39	2.55	9.84	16.53
Trait x forcing	0.45	-1.26	-0.19	1.13	2.04
Trait x chilling	2.07	-0.81	0.96	3.24	4.82
Trait x photoperiod	-0.91	-1.69	-1.22	-0.61	-0.11

References

Boose, E., E. Gould, and M. VanScoy. 2026. Harvard Forest Data Archive: HF300 (v.14). Environmental Data Initiative.

Government of Canada: Environment and natural resources. 2026. Historical Data.

Luedeling, E. 2020. chillR: Statistical Methods for Phenology Analysis in Temperate Fruit Trees. https://CRAN.R-project.org/package=chillR.